TEAM-WORK PROJECT 1: MISSION TO MARS

Deadline: N/A

PART 1 (20 P)

The aim of this project is to assess the feasibility of a mission to Mars, what implies that, rather than focusing on a great accuracy (that an actual mission always requires), the complexity of the mathematical model will be greatly reduced by adopting the following assumptions:

1. Circular heliocentric and coplanar motion of both the Earth and Mars.
2. Patched conic method:
    - The spacecraft will only be under the Earth's gravitational field until it reaches the Sphere of Influence (SOI-ES) of the Earth-Sun system.
    - Upon crossing the SOI-ES and until reaching the Sphere of Influence of the Mars-Sun system (SOI-MS) the spacecraft will only be under the Sun's gravitational field.
    - After crossing the SOI-MS, the spacecraft will only be under the gravitational field of Mars.
3. The heliocentric motion of the spacecraft takes place with no thrust (i.e., coasting), resulting then in a Keplerian orbit.

Since the radii of both spheres of influence are very small compared with the scale of the orbits of the Earth and Mars, the heliocentric coasting flight may be regarded as an elliptic (Hohmann Transfer) orbit. Thus, the heliocentric departure velocity of the spacecraft will given by the sum of the heliocentric velocity of the Earth and the geocentric velocity of the spacecraft upon reaching the SOI-ES. Instead, upon crossing the SOI-MS, the areocentric velocity of the spacecraft will be the difference between the heliocentric velocity of the spacecraft and the heliocentric velocity of Mars.

Using reliable bibliographic references, the following is requested:

1. Find the orbital elements of the heliocentric motion of Earth and Mars, and justify that the assumption of coplanar circular orbits of the Earth and Mars is acceptable for the current analysis of mission feasibility. (5 P)
2. Determine the radii of both spheres of influence SOI-ES and SOI-MS, and justify that the heliocentric Hohmann transfer approach for the coasting phase is reasonable for this mission analysis. (5 P)
3. Determine the relative position between the Earth and Mars (angle between their respective position vectors) at the beginning of the coasting phase, and compute the coasting time of flight, according to the heliocentric Hohmann transfer. (10 P)

PART 2 (40 P)

Consider that the launcher vehicle puts the spacecraft in circular parking orbit around the Earth with a radius of 20000 km, from which the spacecraft performs a low thrust continuous climb up to the sphere of influence of the Earth – Sun system SOI – ES, so as to reach the required velocity increment corresponding to the first "impulse" of the Hohmann Transfer (found in PART 1).

Assuming that the spacecraft is fitted with a certain thruster (probably a nuclear rocket) with a specific impulse $I_{sp} = 900\ s$ and has an initial mass $m_0 = 30000$ kg, the following is requested:

4. Write a code to solve the system of O.D.E of the spacecraft's motion, and find the required thrust, as well as the flight time and propellant mass required, taking the thrust vector always aligned with the velocity vector. (20 P)

Remark: an iterative procedure on the thrust magnitude must be carried out, such that upon crossing the SOI – ES (when the integration ends) the increment of velocity coincides with the first "impulse" of the Hohmann Transfer.

Consider now the approach of the spacecraft to Mars after crossing the Sphere of Influence of the Mars – Sun system SOI – MS. The following is requested:

5. Write a code to solve the system of O.D.E of the motion of the spacecraft, and find the required thrust in order to descend to an altitude of 30000 km above Mars surface, with a circular orbit. (20 P)

Remark: It is recommended to begin from an assumed spacecraft mass in a circular orbit at 30000 km above Mars, and integrate backwards in time (adding mass) in a climb trajectory until reaching the Sphere of Influence of the Mars – Sun system SOI-MS, checking that the modulus of the final velocity roughly coincides with the second impulse of the Hohmann transfer, and the mass of the spacecraft roughly coincides with that of the coasting phase. Two iterative procedures (one inside the other) are then required, one for the spacecraft mass, and (for a given spacecraft mass) another iteration for the thrust magnitude.

PART 3 (40 P)

Assume now that the spacecraft must drop a small satellite in circular orbit at an altitude (above Mars) where its motion is synchronous with the rotational motion of Mars. The orbital plane of the satellite must be gradually changed until becoming coplanar with Mars Equatorial plane, using thrust vectoring, so that the thrust vector forms an angle α with the instantaneous orbit plane, and said angle α changes periodically according to the azimuthal angle φ, and the in-plane component is parallel to the velocity vector:

$$\boldsymbol{F} = F_{\perp}\boldsymbol{u}_{\perp} + F_{||}\boldsymbol{u}_{||} \qquad \boldsymbol{u}_{\perp} = \frac{\boldsymbol{r}\times\boldsymbol{V}}{||\boldsymbol{r}\times\boldsymbol{V}||} \qquad \boldsymbol{u}_{||} = \frac{\boldsymbol{V}}{||\boldsymbol{V}||}$$

$$F_{\perp} = F\sin(\alpha) \qquad F_{||} = F\cos(\alpha) \qquad \alpha = \alpha_{max}\cos\varphi$$

The initial mass of the satellite is 200 kg and is fitted with an ion thruster producing a thrust of 0.2 N and with a specific impulse of 3000 s.

For the question that follows, it is particularly advisable to work in spherical coordinates: radius r, polar angle θ and azimuth φ (see Annex).

6. Write a code to consider both the in-plane and perpendicular-to-plane components of thrust assuming $\alpha_{max} = 60°$, and integrate the O.D.E. system until the orbit is coplanar with Mars Equatorial Plane, with a total change of inclination $\Delta i = 30°$, and the period coincides with the Sidereal rotation period of Mars. (30P)

7. Compute the required time to shift the phase 20º eastwards as fast as possible neglecting the variation of mass, with the same thrust acceleration as at the end of question 6. (10 P)

ANNEX: EQUATIONS OF MOTION IN SPHERICAL COORDINATES

Position vector and velocity:

$$\boldsymbol{r} = r\boldsymbol{u_r} \qquad \boldsymbol{V} = V_r\boldsymbol{u_r} + V_\theta\boldsymbol{u_\theta} + V_\varphi\boldsymbol{u_\varphi}$$

$$\boldsymbol{u_r} = (\sin\theta\cos\varphi\,, \sin\theta\sin\varphi\,, \cos\theta)$$

$$\boldsymbol{u_\theta} = (\cos\theta\cos\varphi\,, \cos\theta\sin\varphi\,, -\sin\theta)$$

$$\boldsymbol{u_\varphi} = (-\sin\varphi\,, \cos\varphi\,, 0)$$

Derivatives of unit vectors:

$$\frac{d\boldsymbol{u_r}}{dt} = \dot\theta\boldsymbol{u_\theta} + \sin\theta\,\dot\varphi\boldsymbol{u_\varphi}$$

$$\frac{d\boldsymbol{u_\theta}}{dt} = \dot\theta(-\sin\theta\cos\varphi\,, -\sin\theta\sin\varphi\,, -\cos\theta) + \dot\varphi\cos\theta\,(-\sin\varphi\,, \cos\varphi\,, 0) =$$

$$= -\dot\theta\boldsymbol{u_r} + \dot\varphi\cos\theta\,\boldsymbol{u_\varphi}$$

$$\frac{d\boldsymbol{u_\varphi}}{dt} = \dot\varphi(-\cos\varphi\,, -\sin\varphi\,, 0) = -\sin\theta\,\dot\varphi\boldsymbol{u_r} - \cos\theta\,\dot\varphi\boldsymbol{u_\theta}$$

Kinematic equations:

$$\frac{d\boldsymbol{r}}{dt} = \dot r\boldsymbol{u_r} + r\frac{d\boldsymbol{u_r}}{dt}$$

Newton's Second Law:

$$\frac{d\boldsymbol{V}}{dt} = \dot V_r\boldsymbol{u_r} + \dot V_\theta\boldsymbol{u_\theta} + \dot V_\varphi\boldsymbol{u_\varphi} + V_r\frac{d\boldsymbol{u_r}}{dt} + V_\theta\frac{d\boldsymbol{u_\theta}}{dt} + V_\varphi\frac{d\boldsymbol{u_\varphi}}{dt} =$$

$$= \left(-\frac{\mu}{r^2} + \frac{F_r}{m}\right)\boldsymbol{u_r} + \frac{F_\theta}{m}\boldsymbol{u_\theta} + \frac{F_\varphi}{m}\boldsymbol{u_\varphi}$$

Conservation of mass:

$$\frac{dm}{dt} = -\dot m$$

Summary of Ordinary Differential Equations:

$$\dot{r} = V_r$$

$$\dot{\theta} = \frac{V_\theta}{r}$$

$$\dot{\varphi} = \frac{V_\varphi}{r \sin\theta}$$

$$\dot{V}_r = -\frac{\mu}{r^2} + \frac{V_\theta^2}{r} + \frac{V_\varphi^2}{r} + \frac{F_r}{m}$$

$$\dot{V}_\theta = -\frac{V_r V_\theta}{r} + \frac{V_\varphi^2}{r \tan\theta} + \frac{F_\theta}{m}$$

$$\dot{V}_\varphi = -\frac{V_r V_\varphi}{r} - \frac{V_\theta V_\varphi}{r \tan\theta} + \frac{F_\varphi}{m}$$

$$\frac{dm}{dt} = -\dot{m}$$